

AI writes, we collaborate—or vice versa? Group strategies for using generative AI in collaborative writing assignments

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ABSTRACT

Generative AI (Gen-AI) tools have gained immense popularity among university students, offering advantages such as reduced time for assignment preparation, enhanced product quality, and accelerated learning. When combined with the benefits of collaborative learning (CL), the group use of Gen-AI holds significant potential. The present study aims to explore how students use Gen-AI during the preparation of a group writing assignment in their academic writing course. Semi-structured interviews were conducted with 10 postgraduate students majoring in AI research, and the data was analysed using thematic analysis. Two independent coders employed QDA Miner Lite software and adhered to Braun and Clarke's (2006) thematic analysis. As a result of analysis, three themes emerged: 1) group strategy: how students establish and implement strategies of collaborative work with Gen-AI application; 2) strategy characteristics: specific Gen-AI functions addressed by students; 3) strategy considerations: perceived benefits, limitations, and other factors affecting group uses of Gen-AI. The findings reveal that students employ Gen-AI either through pre-planned strategies or spontaneous, individually-driven interactions coordinated through shared documents. In both cases, Gen-AI supported not only language and ideation but also integration, helping merge parallel sections into a coherent text and underscoring the need for coordination and verification.

Background

The rapid development of generative artificial intelligence (Gen-AI) since the launch of ChatGPT in November 2022 has significantly impacted higher education [1,2]. In particular, Gen-AI chatbots such as Perplexity, Claude, Gemini, and ChatGPT are now widely used by university students across a range of academic tasks [3,4]. Widespread adoption among university students can be explained by the ability of the chatbots to provide immediate feedback [5,6], assist with assignment preparation [7], and contribute to the development of students' skills and knowledge [8,9]. In the context of academic writing, these characteristics of Gen-AI allow students to complete tasks more efficiently [10] and use it to support idea generation [11], structuring, outlining [12], and revision processes [13].

Recent studies on Gen-AI use show that it can support multiple stages of the writing process, including information search [14], idea generation [15], language refinement [16], paraphrasing and summarizing

[17], as well as drafting and revision [18]. In this context, Gen-AI is conceptualized as a tool or collaborator that assists students in performing specific tasks [12]. Research also highlights variation in how students interact with Gen-AI, showing that AI-generated content may be either revised [19] or directly incorporated into their writing [20]. While these studies demonstrate the multidimensional nature of Gen-AI use, they focus primarily on individual contexts of student–AI interaction. Given the benefits of collaborative learning, including improved critical thinking, knowledge construction, and enhanced learning outcomes [21], the integration of Gen-AI into group writing contexts has the potential to reshape how students collaborate and learn.

In contrast to individual Gen-AI interactions, collaborative writing involves multiple participants engaging in shared writing processes, including brainstorming, drafting, revising, and editing [22]. This process relies on structured coordination and group awareness, including the division of tasks and roles across members and integration of individual contributions into a shared text [23]. Such collaboration requires

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ongoing coordination, negotiation, and alignment of interdependent inputs among participants [24]. Thus, collaborative writing is not simply the sum of individual contributions but a coordinated and interdependent process shaped by group dynamics [24,23]. Some recent studies have begun to explore the use of Gen-AI in collaborative writing contexts, highlighting its role in supporting idea generation, feedback, and coordination [25,26]. However, these studies provide limited insight into how Gen-AI use is organized, distributed, and negotiated among group members. In line with traditional collaborative contexts, Gen-AI use in groups is not limited to interactions between individual students and the technology. AI-generated outputs may become shared resources that affect the work of the group members, shape group decision-making, and affect how responsibility is distributed within the group [12,27]. Consequently, understanding Gen-AI use in collaborative writing requires examining not only what functions Gen-AI performs, but also how its use is coordinated and integrated into collective writing processes at different layers.

This highlights the emerging role of Gen-AI as a collaborator, while the specific nature of this collaboration in group work contexts remains underexplored. Drawing on semi-structured interviews and students' written reflections and group writing documents, the present study explores how Gen-AI use is organized, distributed, and enacted within student groups. The study builds on prior research on individual Gen-AI use [15,14] and collaborative writing [23] by bringing these two strands together and examining Gen-AI use within group-based writing processes. While previous studies have focused on student–AI interaction or human–human collaboration, the present study extends this work by exploring how Gen-AI is integrated into collaborative writing at the group level. The study also extends emerging research on Gen-AI in collaborative contexts [25,11,26] by moving beyond a focus on functional benefits to examine how Gen-AI use is organized, distributed, and negotiated among group members. Beyond describing Gen-AI use, the study aims to deepen understanding of how Gen-AI may reshape collaborative learning. By examining how Gen-AI use is organized, negotiated, and distributed within groups, the study investigates how this technology affects coordination, shared responsibility, and knowledge construction in collaborative writing. Such findings may contribute to a better understanding of how collaborative learning processes unfold when Gen-AI enters the group work.

Thus, considering that the specifics of group Gen-AI use are insufficiently understood, the main objective of the present study is to investigate how Gen-AI use is organized, distributed, and negotiated in collaborative writing contexts.

The research questions the present study seeks answers to are:

RQ1. How do students organize and coordinate the use of Gen-AI within their groups?

RQ2. What types of Gen-AI use characterize students' collaborative writing processes?

RQ3. How do students perceive the benefits, limitations, and implications of Gen-AI use in group work?

Literature review

Strategies for collaborative writing

Collaborative learning (CL) involves students working together in groups, contributing individually to a group endeavour to enhance both their learning outcomes and the quality of the final product [21]. Through interaction, students work toward shared goals, exchange experiences, and educate one another [28]. The impact of CL is particularly pronounced when it not only leads to the achievement of group objectives but also expands the knowledge of individual members [29]. The present study adopts an extension of CL, the computer-supported

collaborative learning (CSCL) framework. CSCL builds on the traditional benefits of CL by incorporating technology to monitor and evaluate student activities while providing feedback to support learning [30].

When writing collaboratively, groups vary in how they organize the work and how responsibilities are distributed. Based on interviews about 22 projects, Posner and Baecker [31] described collaborative writing as a set of activities (e.g., brainstorming, planning, writing, editing, and reviewing) carried about through different writing strategies (e.g., single writer, separate writers/divide-and-conquer, joint writing) and coordinated through different document control methods, with participants taking on roles such as writer, consultant, editor, and reviewer. Lowry et al. [23] extended this framework into a taxonomy by refining the strategy labels (e.g., sequential, parallel writing with horizontal vs. stratified division, reactive writing), elaborating definitions, and adding work modes that capture when and where collaborators write (same vs. different place; synchronous vs. asynchronous), emphasizing that coordination and awareness needs shift across these modes. Yim et al. [32] observed these strategies in practice in empirical classroom research with Google Docs, identifying distinct collaboration styles including “Main Writer” and “Divide and Conquer” as well as more interactive revision-oriented styles. These findings showed that groups can adopt distinct collaboration styles even within the same tool and setting and that these configurations reflect different allocations of writing and editing labour across members [32].

While these studies provide a detailed perspective of how collaborative writing is enacted, they do not account for the role of Gen-AI in group work. However, as an interactive technology, Gen-AI may reshape collaborative work by affecting coordination, role distribution, and interaction among group members.

Strategies for using Gen-AI in individual writing tasks

With the widespread availability of large language models for writing, students have started to use Gen-AI chatbots to help them complete writing assignments. One way to conceptualize this Gen-AI usage is as a collaboration between a student and a chatbot, where students assign the chatbot to perform many of the writing tasks identified by Posner and Baecker [31]. For example, students may assign Gen-AI to assist with abstract tasks such as idea generation, offering fresh perspectives, inspiration, and making the writing process faster, all of which may result in a more enjoyable writing process [15,33]. Gen-AI also facilitates such processes as paraphrasing [19], summarizing, and categorising [17] of the texts that students then incorporate in their writing. Gen-AI may also create an initial structure or outline of the task for students to build upon, change, add their own ideas to, and logically strengthen [33]. Gen-AI provides substantial assistance in refining and editing the final texts produced by students by offering targeted feedback with explanations and enhancing overall quality [15]. As for less abstract Gen-AI applications, Gen-AI is able to assist with routine, time-consuming activities like retrieving information on the topic [33, 19], formatting individual writing, and managing citation [15]. It is also widely used for such mechanical tasks as error correction [19]. Gen-AI is capable of improving writing language by creating stronger sentences, enhancing grammar, and ensuring accurate vocabulary use [19]. Gen-AI is also employed for translation [17].

While these studies demonstrate the ways in which Gen-AI supports individual writing, they conceptualize Gen-AI use as an interaction between one student and the system. It remains unclear how these functions translate into collaborative contexts, where writing involves coordination, role distribution, and shared responsibility among the group members.

Strategies for using Gen-AI for collaborative writing

Studies highlight Gen-AI's role in enhancing creativity during group

writing, particularly in brainstorming, where it enriches ideas initially generated without its use [11,26]. Gen-AI significantly reduces the time required for idea generation in group contexts [11] and supports creating argumentation [25]. Additionally, it facilitates topic exploration by searching and synthesizing information [34,26]. Students find Gen-AI chatbots effective for answering topic-related questions and clarifying unclear points, although concerns about accuracy remain [35, 36]. These tools also improve the language quality of group writing through grammar, style, and spelling corrections, synonym suggestions, and language refinement [25,26]. Furthermore, they provide preliminary assessments of writing and constructive feedback for improvement [26]. A notable challenge in group work with Gen-AI is synthesizing individually completed sections into a cohesive final text [34]. However, existing research focuses on the functions that Gen-AI performs in group contexts rather than on how its use is organized and coordinated among group members. The way Gen-AI is integrated into collaborative writing processes at the group level remains unclear.

Overall, while a large body of research describes how students collaborate with other students when writing in groups and a large number of studies have examined how students work together with Gen-AI chatbots to complete their individual assignments, there is a lack of research examining how student groups interact with Gen-AI chatbots when writing collaboratively.

Materials & methods

Learning context and procedures

The present study was conducted at a large Korean university and involved graduate students majoring in AI research enrolled in a scientific writing course taught in English. Throughout the course, students worked in fixed groups of 4–5 to complete a series of collaborative scientific writing assignments, including the writing of the Introduction and Discussion sections of an academic paper. They were encouraged to use Gen-AI in any form they deemed necessary to support their work. At the end of the course, all 143 students of the course received an email invitation to participate in interviews designed to explore their use of Gen-AI during group tasks. Participation was voluntary, and 10 students agreed to take part. The interviews were conducted online, with each lasting approximately 30 min. The sessions were recorded and transcribed. The interview included questions about establishing a group strategy for using Gen-AI, students’ individual approaches to applying this technology within the group, and the perceived efficiency and effectiveness of their chosen Gen-AI uses. To triangulate the interview data, self-reflection notes written by the interviewed students after each task were analysed. These notes detailed how Gen-AI was employed in each task. In order to supplement the interview data, students’ written responses on how exactly they used Gen-AI were analysed. Out of 143 students, only the answers of the interview participants were considered.

Participants

The sample consisted of 10 university students aged 23 to 39 years ($M = 27.7$, $SD = 5.4$), all enrolled in the Graduate School of AI. This population of students was specifically targeted for participation in the study due to their knowledge of and familiarity with the use of Gen-AI tools. As shown in Table 1, the participants included 3 females and 7 males. Of these, 7 were enrolled in master’s programmes, and 3 in a doctoral programme.

Analysis

The data were analysed using thematic analysis following Braun and Clarke [37]. This approach was chosen as it allows for the systematic identification of repeated patterns in the data while remaining flexible

Table 1
Interview participants’ demographic information.

Participant	Male/Female	Age	Level of Education
I 1	M	23	Master’s
I 2	M	24	Master’s
I 3	M	25	Doctoral
I 4	F	36	Doctoral
I 5	M	28	Master’s
I 6	M	26	Master’s
I 7	F	25	Master’s
I 8	M	39	Master’s
I 9	M	26	Doctoral
I 10	F	25	Master’s

in capturing emergent aspects of the phenomenon. Two coders coded the data in QDA Miner Lite software, and in case of disagreement, the third coder, a research team member, was consulted. The analysis was conducted in four stages, which are described below.

The analysis followed a predominantly inductive approach, as no predefined coding framework was used. Initial codes were generated directly from the interview data and subsequently refined through the four stages of analysis described below. During the second stage, the refinement of the coding scheme was informed by relevant literature on Gen-AI use in academic writing. Through an iterative process of coding, discussion, refinement, and validation, the initial codes were consolidated into three overarching themes (Group Strategy, Strategy Characteristics, and Strategy Considerations) and associated codes.

Stage 1. Familiarization and initial coding. This stage focused on familiarization with the data and the development of a preliminary coding scheme. The researchers independently read all interview transcripts and conducted initial coding of the first transcript as a pilot phase. Each coder generated an initial set of codes based on their interpretation of the data. The initial codes included:

- group strategy of Gen-AI use
- individual strategy of Gen-AI use
- benefits of Gen-AI use
- efficiency vs. effectiveness of Gen-AI use
- modification of Gen-AI outputs.

A meeting was then held to compare and discuss these initial codes, resulting in a preliminary coding scheme (see Fig. 1).

Stage 2. Refinement of coding scheme and initial theme development.

At this stage, transcripts from interviews 2–4 were coded to examine whether patterns identified in the initial coding were consistent across the data. The analysis focused on refining and expanding the preliminary coding scheme. A follow-up meeting was held during which the coding scheme was discussed and further developed (see Fig. 2). Three overarching themes began to emerge:

- (1) *group strategy*, including codes such as establishing strategy and individual or constituent strategies within the group;
- (2) *strategy characteristics*, including types of Gen-AI use (e.g., language support, idea generation), sources of initial drafts (human vs. AI), and perceptions of competence;
- (3) *strategy considerations*, including risks and benefits of using AI.

The refinement of the coding scheme was informed by prior research on Gen-AI use in academic writing, particularly distinctions between technical [14,16] and non-technical [18,15,17] categories of tasks that Gen-AI can assist with.

Stage 3. Theme refinement and consolidation

The third stage involved analysing interviews 5–7 with the aim of refining and consolidating the thematic structure. Particular attention

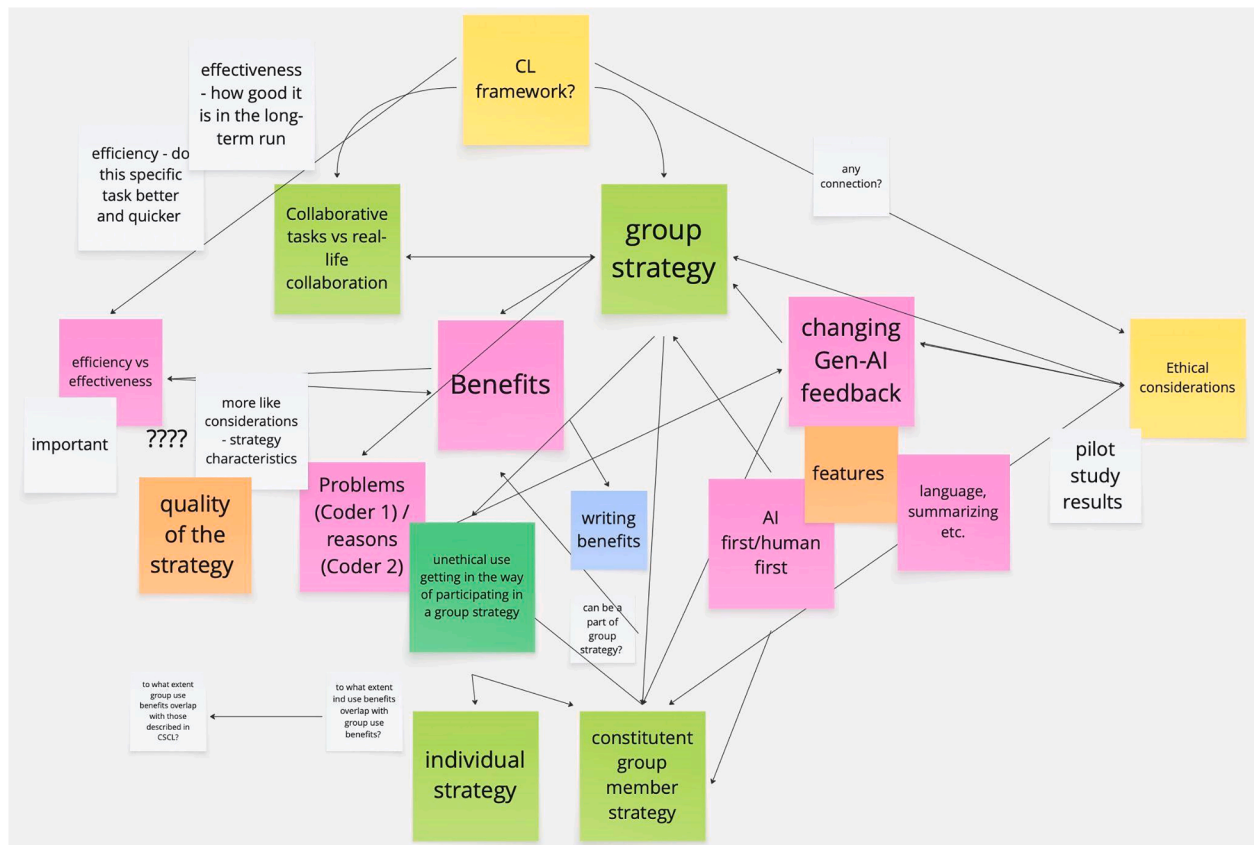


Fig. 1. Development of the preliminary coding scheme.

was paid to ensuring that the identified themes were consistently represented across the data and captured meaningful patterns in students' use of Gen-AI. A follow-up meeting was held to review and refine the coding scheme. As a result, the structure of themes and associated codes was clarified. Within the group strategy theme, only codes directly related to the formation and enactment of group and constituent strategies were retained. Within the strategy characteristics theme, specific uses of Gen-AI were reorganized into broader categories of technical and non-technical use, and additional codes were introduced to capture stages of use and the distinction between AI-generated and human-generated initial drafts. The code related to modification of Gen-AI outputs was retained. Within the strategy considerations theme, additional codes were introduced to differentiate the impact of Gen-AI use on the learner, the written product, and the learning process.

Stage 4. Validation of coding scheme

In the final stage, interviews 8–10 were coded using the refined coding scheme. This stage aimed to confirm that the coding framework was applicable across the data and that no new significant patterns emerged. A final meeting was held to review the coded data and ensure agreement among researchers. This process confirmed the stability of the coding scheme. The final coding framework is presented in Table 2.

In order to triangulate our findings and validate the coders' interpretations of the interview data, we used two additional data sources, students' written reflections on their Gen-AI use and their group documents with the Introduction and Discussion tasks. Interview transcripts served as the primary source for theme development and coding. Once the interviews had been coded and illustrative quotations had been identified, the researchers analysed students' written reflections on their Gen-AI use during the tasks to elaborate on patterns identified in the interviews: make them more detailed and illustrative. Finally, group writing documents were examined to determine whether the reported

uses of Gen-AI were reflected in students' actual writing practices and contributions.

Figs. 1 and 2, together with Table 2, document the development and refinement of the coding framework and illustrate the analytic process. The remaining figures (Figs. 3,4,5,6,7,8,9,10,11) are presented in the Results section and illustrate the findings of the study.

Results

The analysis revealed three interconnected themes that explain how students used Gen-AI in collaborative writing tasks (Fig. 3).

The first theme, "Group strategy", describes how students organized and coordinated the use of Gen-AI within their groups, including whether Gen-AI use was planned collectively or left to individual group members. The second theme, "Strategy characteristics", captures the specific ways Gen-AI was used during the writing process, including technical and non-technical applications. The third theme, "Strategy considerations", reflects students' perceptions of the benefits, limitations, risks, and ethical implications associated with group use of Gen-AI. Together, these three themes provide a multi-layered understanding of how Gen-AI is integrated into collaborative writing, from group-level organization to specific uses and students' reflections on these practices.

RQ1. How do students organize and coordinate the use of Gen-AI within their groups?

Theme 1. "Group Strategy"

As reflected in Fig. 4, the "Group Strategy" theme includes three large codes: "Establishing Group Strategy," "Realization of a Group Strategy," and "Constituent Strategy." The figure illustrates the conceptual structure of the theme and its subthemes, while detailed

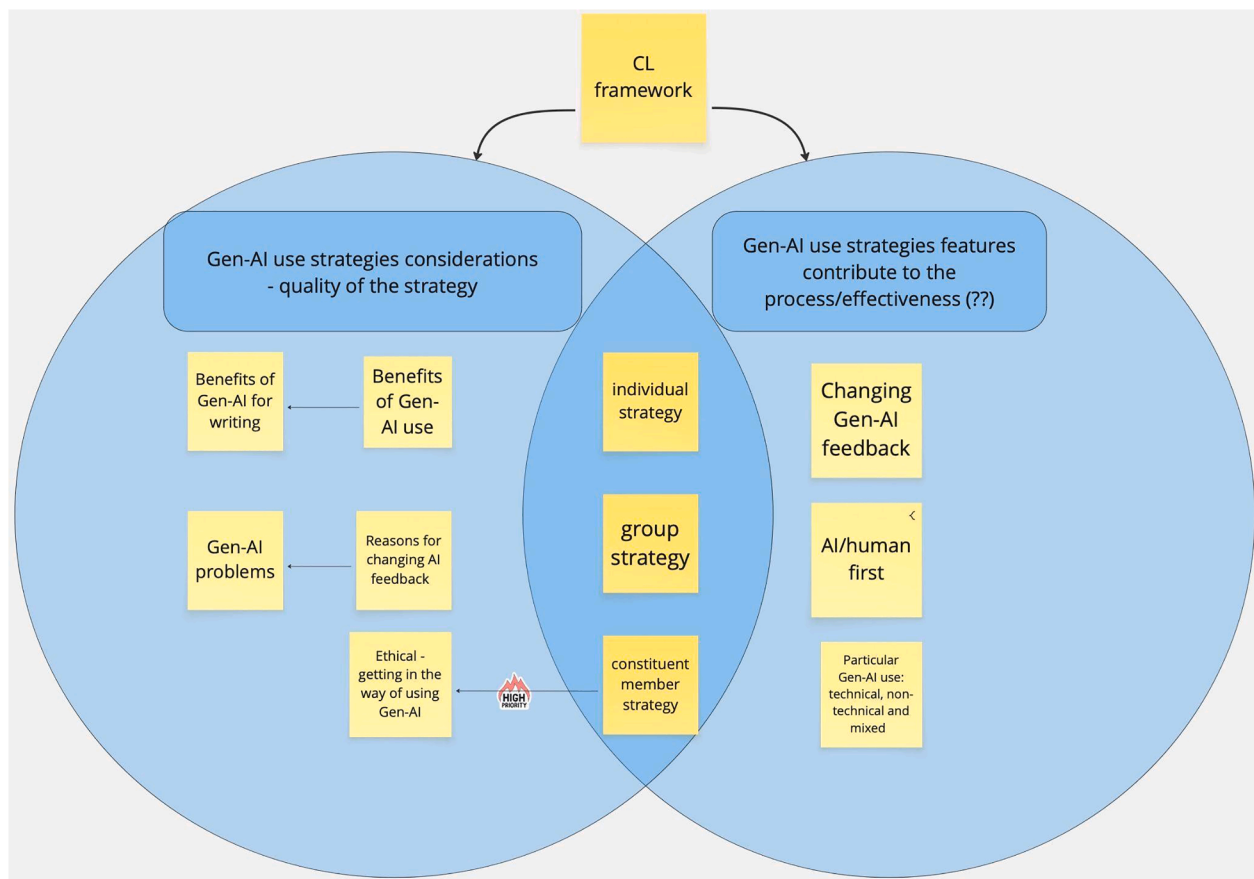


Fig. 2. Refinement of the coding scheme.

descriptions of each subtheme are provided in subsequent figures and sections. This theme captures the organizational dimension of collaborative Gen-AI use, highlighting how students coordinated, distributed, and negotiated the role of Gen-AI within their groups before and during the writing process.

Establishing a group strategy

One way to approach a group task, as can be seen in Fig. 5, is to establish a group strategy, where students discuss how to employ Gen-AI before starting their work, including specific steps such as choosing a topic and creating an outline: “in that case my teammates say that let’s generate the draft for the introduction with ChatGPT. So, we used ChatGPT with my teammates” (12).

Sometimes, it was a group decision; other times, one student took the lead, and the others followed: “So usually there was one student that initiated all this writing, like assigning parts and telling what like, starting the discussion about the topic you should write on” (15).

The other way to work as a group is when students did not discuss Gen-AI use in advance; instead, the final outcome consisted of individual contributions. In this mode of work, the group members decided how to use Gen-AI independently in their own parts within the task: “So, the whole decision was really dependent on the people” (15). An important point is that there are certain limitations students consider as a group even if they choose to go with an “open” approach to using Gen-AI: “So whatever they want, they can, like, do anything. And, like, except generating or writing false information” (18). The first reason students gave for not having a pre-decided group strategy was the availability of a virtual space that let them see what others were doing: “We didn’t, we didn’t explicitly set a policy, but it was mostly like because we can see who’s writing what in the Google Docs” (14). Other reasons included the failure to encourage others

to create a joint strategy and a lack of time to do so. These findings suggest that group coordination of Gen-AI use existed on a continuum ranging from explicitly negotiated strategies to implicit and largely unspoken practices that emerged during collaboration.

It is interesting that during the interview a student (15) said that the group members had no group strategy, with one group member assigning specific writing tasks to group members. However, in written responses the same student actually described the group strategy that was used: “We first generated an initial draft by providing ChatGPT with the same instructions from this document and five criteria that the Professor mentioned. This way, we could individually work on rewriting paragraphs while watching others’ progress, which helped us write more coherently and faster” (15). This is a contradiction unless the student was unaware of the group plan and, therefore, was not consciously involved in it. This discrepancy suggests that group strategies were not always equally visible to all members and that some collaborative practices may have been perceived as individual rather than collective decisions.

Realization of a group strategy

As shown in Fig. 6, when students had a pre-decided group strategy for using Gen-AI, it usually involved producing a draft of the writing task first: “Like all the group assignments, we tried to use the...to write like the...like the first draft of our work” (11).

The first draft, if produced by Gen-AI, was used not only as a writing sample but also as a way for students to familiarize themselves with the topic and generate ideas: “the first draft of our work. It usually gives like a general perspective of the topic, so. We use that to first generate some high-level ideas on that topic” (11). The group document associated with this student provides textual evidence that the AI-generated draft was substantially revised: only a limited portion of the text originates from AI

Table 2
Themes, codes and code descriptions developed in the process of analysis.

Theme	Code	Code description
Theme 1 "Group strategy"	Establishing group strategy	Whether the group discussed how to use Gen-AI
	Realization of a group strategy	The way Gen-AI was used by students as a group
	Constituent strategy - personal	How students used Gen-AI individually within the group strategy
	Constituent strategy - others	How students perceive individual use of Gen-AI within a group strategy by others
Theme 2 "Strategy characteristics"	Specific Gen-AI use - technical	Language-related uses, typos, error, translations and similar functions
	Specific Gen-AI use - non-technical	Abstract uses like summarizing, paraphrasing, editing
	Specific Gen-AI use - mixed	Combinations of technical and non-technical Gen-AI uses
	Gen-AI first/human first	Who produced the first draft - student(s) or technology
	Changing Gen-AI feedback	Whether students changed Gen-AI feedback and if so, how and how much
Theme 3 "Strategy considerations"	Benefits of Gen-AI use - the document	How Gen-AI was beneficial for group writing that students produced
	Benefits of Gen-AI use - the learner	How Gen-AI was beneficial for learners in the long-term: skill, new knowledge and similar
	Benefits of Gen-AI use - the process	How Gen-AI was beneficial for the process of group work
	Limitations of Gen-AI	Downsides of Gen-AI use
	Harm of Gen-AI - present task	Downsides of Gen-AI use for the writing assignments
	Harm of Gen-AI - long-term	Downsides of Gen-AI use for learners and learning in long-term perspective
	Ethics in group work context	Ethical aspects of Gen-AI use that impacted group of constituent strategies

(highlighted in yellow), while multiple font colours indicate extensive turn-taking among group members (see Fig. 7).

In some cases, students asked Gen-AI to provide several drafts and selected the one with the best quality: *"so we've made the two output generated from...the Gen-AI stuff and we've just choose one thing as our main paper for the last assignment so because the one of the thing was having some low quality than the than the others"* (16). In both approaches, students worked under their individual strategies, regardless of whether they

discussed Gen-AI use as a group. In the end, students wrote sections in parallel and then used a group-specific Gen-AI merge step to integrate and revise them into one cohesive text, aligning with Lowry et al.'s [23] taxonomy of parallel writing with horizontal division of labour followed by integration: *"usually I or the other student had to, like, collect all the responses and, like, make or glue them into, like, combined paragraphs. So, for that, Gen-AI is really good"* (13). The reason for using Gen-AI in this way was that *"if you merge them, sometimes the information overlaps or sometimes, uh, it doesn't connect"* (18). In this case, Gen-AI took on the role of an editorial integrator during the merge step. Rather than serving only as a source of content, Gen-AI was used to coordinate and integrate individual contributions into a coherent group product.

Constituent strategy

Students interacted with Gen-AI individually, either as part of a pre-decided group strategy, where they all agreed on how to use it and were assigned their parts, or by following a strategy where the decision on how to apply Gen-AI was not discussed and was left "open" to the needs of individuals. Students used Gen-AI individually to address areas where they struggled: *"I just used AI, uh, Gen-AI for the, uh, like some supporting some my, some, how can I say, some weakness"* (17). Gen-AI is reported to be particularly helpful in this regard, as it helps address the weaknesses of each group member, thereby evening their level with the level of other group members and improving group dynamics: *"because different people have different levels of English writing skills or understanding of Gen-AI or understanding of the topic. So it seemed actually better to have an AI embedded in our content"* (15).

Generally, the interviewees perceived that their groupmates were applying Gen-AI in the same manner as themselves: *"but I think most of them use them to revise their writing, I think most of them. And I think that's what I observed. And yeah, so kind of similar, sometimes they use Gen-AI to brainstorm as well. And yeah, so I think for the group work, I think like that"* (18). However, those who used Gen-AI for technical tasks, such as grammar assistance, perceived their groupmates' abilities as higher than their own and assumed they were using it for more conceptual writing elements, like content development. These findings suggest that individual uses of Gen-AI remain closely tied to learners' perceived strengths and weaknesses, even when writing is organized as a collaborative activity. The main components of the constituent strategy are illustrated in Fig. 8.

Individual interactions with Gen-AI consist of a number of particular functions, which will be described under Theme 2.

RQ2. *What types of Gen-AI use characterize students' collaborative writing processes?*

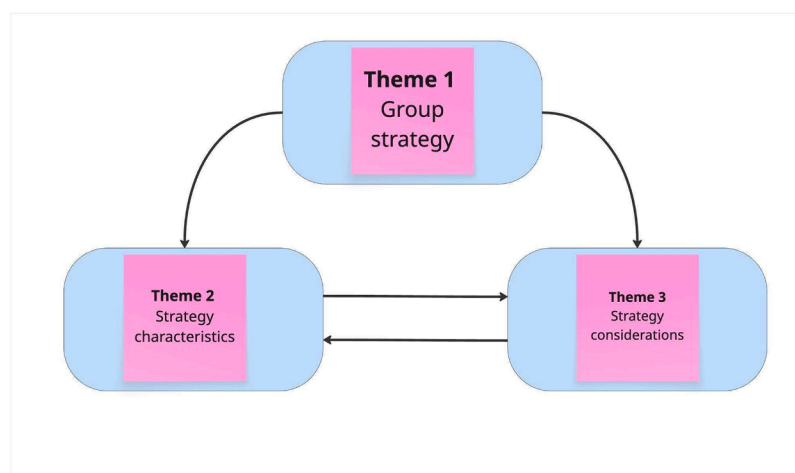


Fig. 3. Gen-AI use themes in collaborative writing.

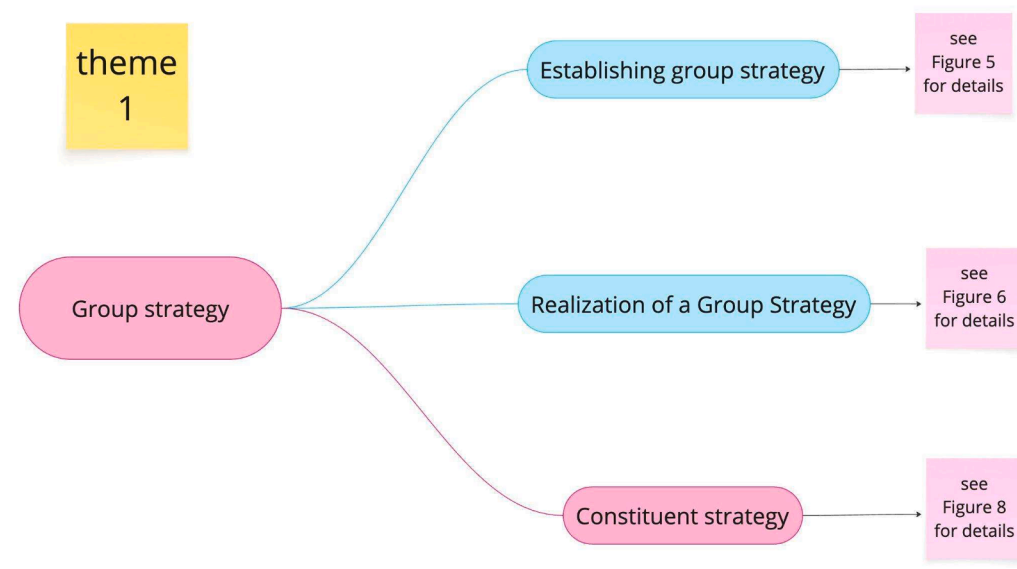


Fig. 4. "Group Strategy" theme and related subthemes.

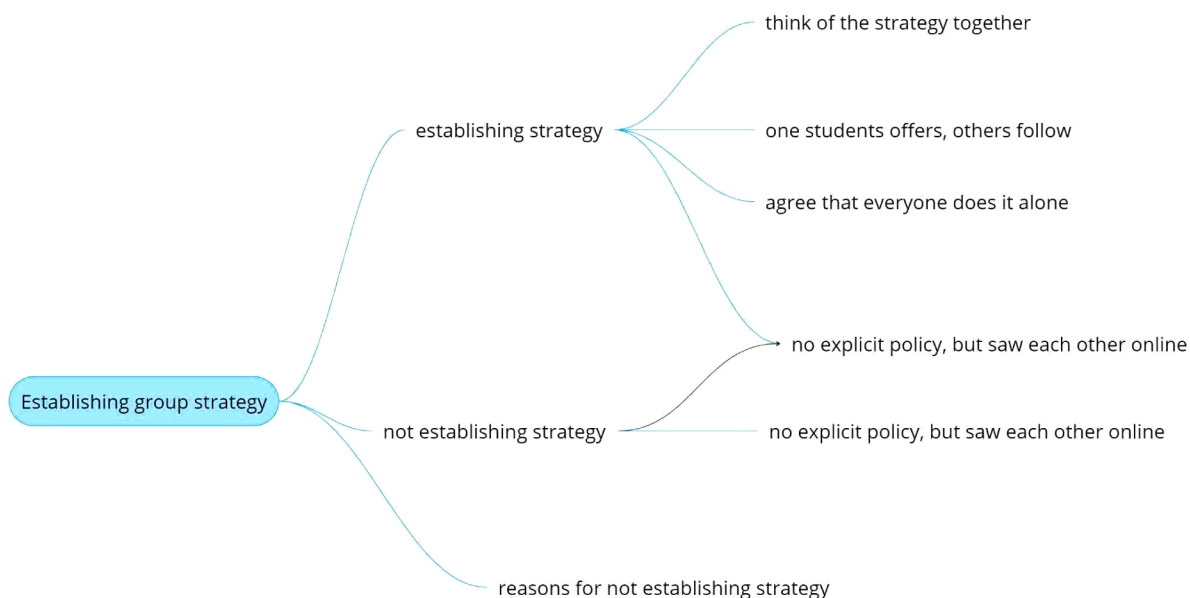


Fig. 5. "Establishing a group strategy" subtheme and related codes.

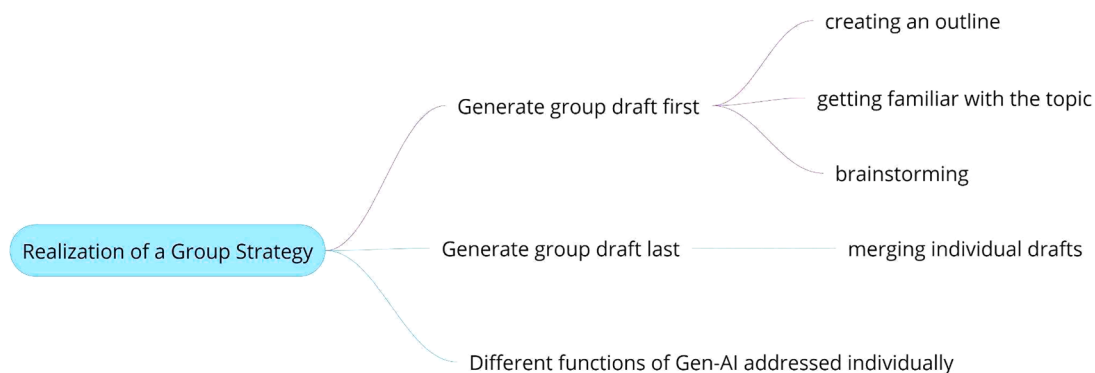


Fig. 6. "Realization of a group strategy" subtheme and related codes.

While instant food, such as ramen, offers convenience, its nutritional value is notably deficient. Historically, ramen was promoted in Korea during food shortages to enhance wheat consumption [3]. This unbalanced composition not only leads to direct health issues, such as thiamine deficiency [1], but also perpetuates negative food habits overall [5].

Nevertheless, the main reason KAIST students buy ramen is for time efficiency. Students already know that healthy food is better, so if healthy food is still slow or not easily cooked, it cannot serve as an alternative to instant food. Furthermore, Healthy foods are difficult to sell in stores and require cooking, making them difficult to be consumed late at night. Considering that instant food is often consumed in places other than the cafeteria, such as dormitories, it is hard to effectively reduce instant food consumption.

Fig. 7. Revision of AI-generated draft and collaborative editing in group writing task.

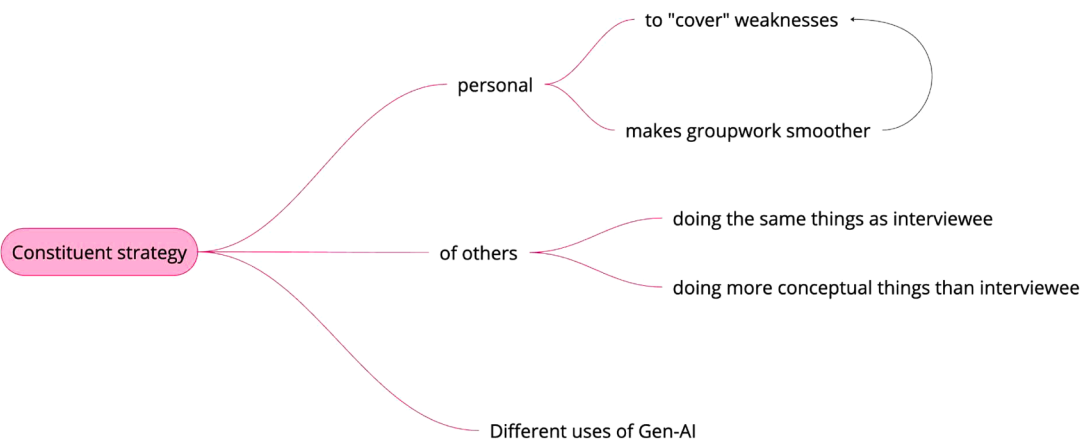


Fig. 8. “Constituent strategy” subtheme and related codes.

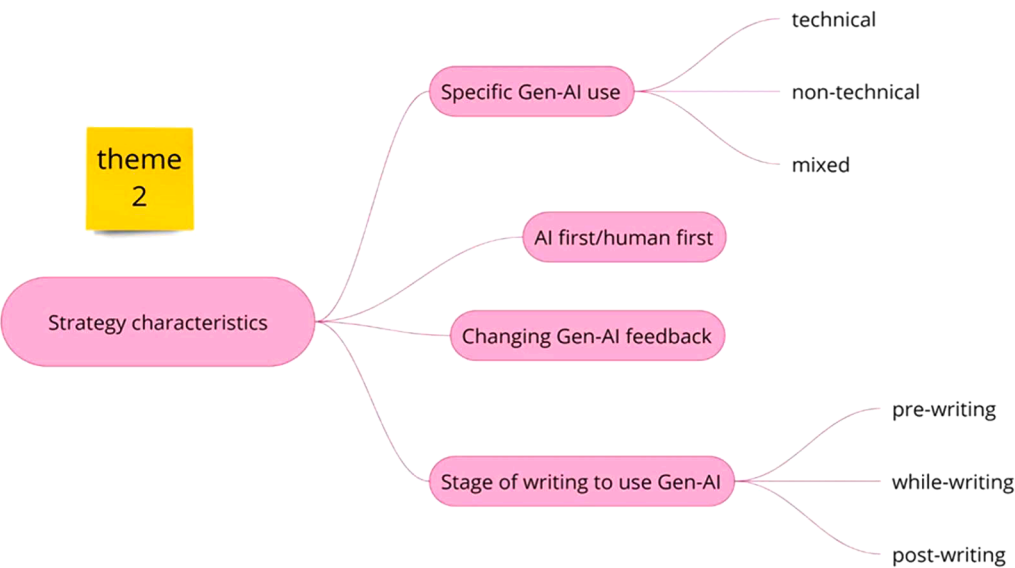


Fig. 9. “Strategy characteristics” theme and related codes.

Theme 2. “Strategy Characteristics”

As illustrated in Fig. 9, this theme expands upon elements of the “Group Strategy” and “Constituent Strategy” discussed earlier, focusing on the specific features of both strategies employed during group work.

It includes such codes as “Categories of Gen-AI Use,” “AI First vs. Human First,” “Stage of Writing,” and “Changes Made.” While Theme 1 focuses on how Gen-AI use was organized, this theme focuses on how Gen-AI was actually enacted during collaborative writing through different

The objective of the research was to develop a robot that can replace a student in a classroom setting. Our findings indicate that our robot, the iStooge, achieves superior attendance scores and higher average grades than other machines documented in literature. The iStooge significantly outperforms the Hubo and the T9000 over all subjects, thereby reducing the gap between robotic and human academic performance.

In terms of attendance, we find that the capability to flexibly respond to impromptu queries by the instructor is crucial for demonstrating attentiveness during class participation, leading to higher probability of receiving attendance credits, as shown in Table 1. In contrast to mathematical queries (row 3) that can be answered without context based on logical processing, responding to materials during Engineering or English lectures (rows 1,2) demands an understanding of the underlying context.

Fig. 10. Illustration of individual work on the group writing task.

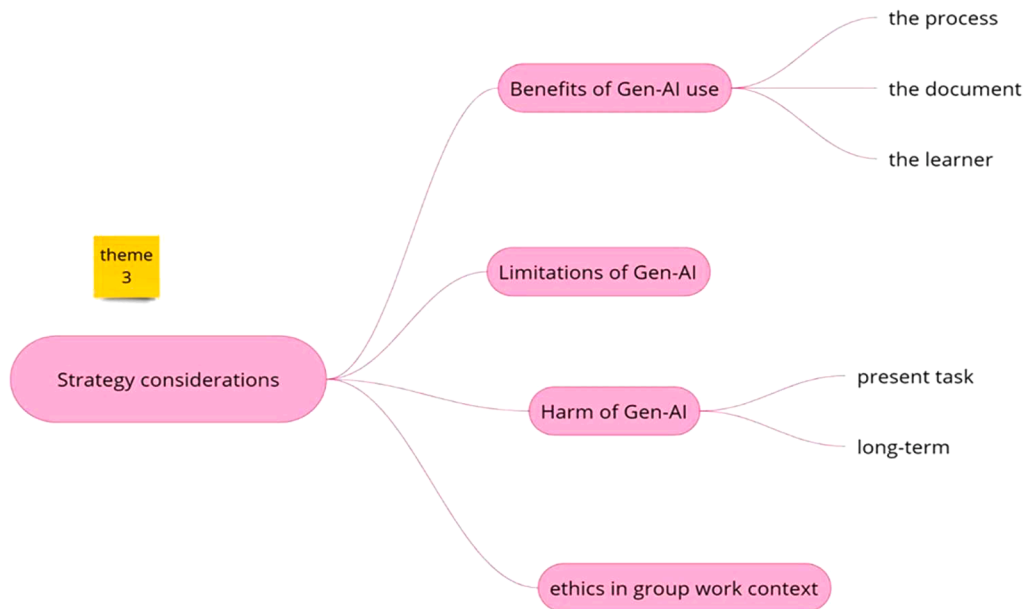


Fig. 11. "Strategy considerations" theme and related codes.

functions, stages, and patterns of interaction.

Specific use of Gen-AI

This code captures the *specific* ways that students used Gen-AI in their constituent tasks within the group task. The uses were categorized as if Gen-AI were a groupmate, with students asking it for help with specific tasks, essentially determining what kind of thinking function they would apply in that situation – technical or non-technical (abstract). Students employed a variety of technical uses of Gen-AI.

For example, they utilized this technology to help them fix grammar: "And, moreover, we used Gen-AI as a grammar corrector, because ...English, is not our natural language" (I2); to look for appropriate words or phrases: "maybe if I cannot, uh, come up with some good, sentences or the appropriate the keywords" (I7). Another popular function was improving their writing style: "personally, I used Gen-AI to hip up the tone phenomenon, then I just, like, wanted to pinpoint to, like, right expression" (I3). The textual evidence shows that the student (I3) highlighted AI-assisted text in yellow, including expressions such as "considerable room for improvement" and "yielding promising outcomes." This illustrates how

stylistic enhancement ("punching up" the text), as described in the interview, was enacted in practice (See Appendix A, Fig. A1).

Students described Gen-AI as being used to enhance the text coherence: "we also checked the whole the coherence of the whole writing through ChatGPT" (I4). Sometimes Gen-AI was used for translation: "Then I would type in that in Korean to ChatGPT, then ChatGP gave me some complete English sentence, so I will use it" (I2). Textual evidence demonstrates that in the group document, this student (I2) used Gen-AI to generate the entire sentence, which can be interpreted as an instance of using Gen-AI for translation (See Appendix A, Fig. A2).

Students also asked Gen-AI to perform a variety of non-technical tasks. This included brainstorming: "So I give the all the information to the AI, and I just want to do some brainstorming. For example, how can I, how should I explain the table? What are the key findings in the table?" (I8) and developing basic ideas of the students: "So, I try to make my own some any some um, thinking, some base, some basic thinking, and I asked to Gen-AI for the, my, like, I gave some my circumstances for the writing, what is the good sentences for the text situations." (I7). Another popular non-technical use of Gen-AI was connected with the drafts of the task. For example, Gen-AI assisted the writing process by producing the first draft

of the text: “So basically, the Gen-AI wrote the first draft, and then we individually added our own ideas and refined the sentences into our own language” (I5). Another task was to merge the drafts produced by the group members individually into one product. Usually, for this, the most experienced Gen-AI user in the group was chosen: “I think it was mostly me and another person. I don’t remember the name, but because we were more used to using that ChatGPT in our like, everyday, uh, assignments or anything” (I4).

Closer to the end of writing process, Gen-AI was asked to act as a “judge” or assessor of the product, proofreading what had been written: “I also, uh, determined that whether oh, it’s good or it’s not, or which it need to be something fixed or something like that” (I6). Evidence from the group document shows that most of this student’s (I6) (green font) text is highlighted in yellow, indicating that it was likely processed or revised using Gen-AI (See Appendix A, Fig. A3).

Other students similarly used Gen-AI for editing their writing: “I first write my first draft, and then I ask AI to revise my draft to be like professional one. So my English is kind of like rough, like not good, like writing that level. So I ask AI to revise my writing.” (I8) or grading: “after I revised my, uh, the writing, I ask feedback from the AI...like professional reviewer. And I ask something like that. And I also provide the rubric” (I8).

There were cases when students, in their writing, employed both technical functions of Gen-AI like, for example, translation, and non-technical, like brainstorming: “Then we get the some first draft from the GPT, and then we fixed the that draft and develop some...develop it. And, moreover, we used Gen-AI as a grammar corrector” (I2). Together, these findings indicate that students employed Gen-AI as a multifunctional collaborator capable of supporting both surface-level language work and higher-level conceptual aspects of writing.

It is important to highlight that the same group might have used Gen-AI differently depending on the task. For example, while working on the Introduction task, the work with Gen-AI was mostly language-related. However, while working on the Discussion task, the whole group tended to use Gen-AI for brainstorming, while the student (I9) criticised this function of Gen-AI and decided to employ Gen-AI only to work with the language: “In writing, I like to first quickly write something down in casual language, and then ask GPT-4 to rewrite it in academic style. I can then copy the phrases and formulations that I like to my own text” (I9). This indicates a possibility that the students were not aware of what the other group members were doing, which can be confirmed by the following quote: “I mean, it’s a writing assignment, right? So, it’s just like, I only cared about my paragraph. But the other paragraphs, this is like, they do their thing, I do my thing” (I9). The structure of the group document in which this student (I9) participated supports this statement (see Fig. 10).

AI first vs human first

Another perspective on the use of Gen-AI is who produced the first draft: humans or Gen-AI. This concept is closely connected with “Group strategy,” as the group member who made the first version of the text was connected to the type of strategy the group selected: pre-decided or open. This was confirmed by individual students who said they did not ask Gen-AI to produce the entire text if they worked alone in general, not just for this task: “personally, I don’t make Gen-AI, like, plain answer, like the questions, or I just don’t give it like questions without like my original responses. So, I first, like, I tried to write off something first, and I just, like, use it as an input with like my questions relevant to that” (I3). Using Gen-AI to make the first draft was easier for a group because students needed a consensus on their ideas and further directions before they started writing: “So, it was kind of hard to, like gather all those ideas from each person and write it into a full draft. So, once we got the whole topic of our group, it was easier for Gen-AI to create a first draft of it and then refine the strap into our own words.” (I5).

Changes made

This code reflects the extent of modifications applied to the final document after incorporating Gen-AI. It also includes additional considerations such as the reasons for edits and their impact on the group’s workflow and outcomes. Students indicated that they could not use AI-generated feedback directly, as there might be some flaws: “But we cannot use that directly, because as like generative AIs produce a lot of fake answers like they have like hallucination, and that kind of problem” (I1), or AI-produced information could be outdated: “Also like there are also another problem with generative AI having answers that are not up to date” (I1). Students admitted that copy-pasting AI outputs with no changes hinders learning process, and that effortful interaction with this technology is needed in order to learn: “So, if I really want to learn something, then I should not just, like, get the output from Gen-AI directly and use it like as like my it’s my own response. I think it’d be more helpful to ask it why did you make the changes to my original paragraphs” (I3). Such answers indicate that Gen-AI is a capable contributor to the group work but that there are risks to offloading important tasks to it.

RQ3. How do students perceive the benefits, limitations, and implications of Gen-AI use in group work?

Theme 3. “Strategy Considerations”

This theme encompasses considerations behind particular Gen-AI uses on group and individual (constituent) levels. As can be seen in Fig. 11, the theme includes the following codes: “Benefits of Gen-AI Use,” “Limitations of Gen-AI Use,” “Harm of Gen-AI Use,” and “Ethics.” Unlike the previous themes, which focus on organizational and behavioural aspects of Gen-AI use, this theme captures students’ evaluations and perceptions of these practices, including perceived benefits, limitations, risks, and ethical concerns.

Benefits of Gen-AI use

When discussing the benefits of Gen-AI, its advantages for the process, the outcomes of group work, and long-term development were highlighted by the students.

Improving the Writing Process. As for the process, it was noted that Gen-AI made group work easier by minimizing the need for members to check each other’s work, thereby helping them to avoid redundancies: “So, it’s definitely efficient that, even after my full writing, I don’t have to do like, grammar checks one by one, and also checking the maybe the coherence or, like, the general flow became much easier after you started using ChatGPT.” (I4). Gen-AI simplified the writing process by eliminating language-related challenges, enabling students to focus on the content: “now with Gen-AI, I can just write whatever awkwardness I want, and then you use the Gen-AI to fix it, and it sort of, like, relieves a lot of the burden” (I9). Gen-AI language support also helped students overcome writer’s block: “it helps me speed up the writing process by the fact that I don’t postpone my work to tomorrow, because I hate writing” (I9). Using Gen-AI also helped students to do the task faster: “So, um, uh, one of my answers is that the main strength of the Gen-AI is just we can do more things in short times. For example, we need to write so many words for our articles. But by using Gen-AI, we can get some, uh, quite normal or high quality from the Gen-AI in just one or two seconds” (I6). As for the resulting product, Gen-AI assisted with improving formal characteristics of the final group: “I think this kind of AI-generated sentences are much more eloquent, grammatically correct and concise” (I5).

Enhancing Skills and Broadening Knowledge. The benefits of Gen-AI for students’ long-term development, according to the students, included enhancing their skills and broadening knowledge beyond the task completion. By interacting with AI-produced text, students familiarized themselves with standards of academic writing, which they could transfer to the other tasks: “so it really helps me, really teach me how I should gather my thoughts into a concise sentence. So, they create example

sentences, and I, and I get repeatedly exposed to kind of sentences. So, I think I became a kind of learned more natural sounding sentences or so.” (I5). They also acquired academic writing terminology and learned how to apply it to the real texts. Furthermore, working with Gen-AI provided students with an opportunity to identify their areas for improvement: “So I can see which part is, uh, has some weakness on my writing” (I8).

Contradictory Accounts of the Benefits of Gen-AI Use. When the interview answers are compared to the written responses on Gen-AI use, some discrepancies were revealed. For example, during the interview, one student (I3) described using Gen-AI in engaged and learning-oriented way, explaining that they first produced their own draft and then used it as input for Gen-AI to refine and expand. The student stated that this approach allowed him/her to remain involved in the writing process and not to rely on Gen-AI excessively. However, in written responses, the same student reflected that Gen-AI was actually used to make the process of writing faster: *“I think that my use of AI was quite effective, as I could make it quickly produce useful insights out of tables, which in turn helped me to sharpen my ideas and sentences. I could also free-write without worrying too much about grammar and minor details then correct them afterwards” (I3) and easier: “I believe that using AI for this type of timed writing is extremely helpful, as it helps me overcome issues like writer’s block or the tip-of-the-tongue phenomenon. It allows me to write without hesitation and reduces my concerns about grammatical errors” (I3).* Thus, even though it was noted that Gen-AI is beneficial for group writing, the real motivation behind its use remains more complex than it is articulated by students.

Harm of Gen-AI use

Participants noted that reliance on Gen-AI could negatively impact development of their writing skills over time: *“but these days, uh, since I’m using it also to, like, brainstorming, I, usually if I think something, I ask like ChatGPT. So I’m becoming a bit lazy these days, not think too much or not writing” (I8).* These concerns extend to the potential impact of Gen-AI on students’ cognitive engagement in the writing process. Students expressed the view that frequent reliance on Gen-AI may reduce their need to engage in complex thinking, which includes generating ideas and developing arguments. They suggested that such reliance could lead to the offloading of thinking processes to the technology, which may be detrimental to meaningful learning. Students also noted that characteristics often perceived as strengths of Gen-AI, such as convenience and efficiency, may encourage this offloading and negatively affect long-term skill development.

Limitations of Gen-AI use

Students highlighted that they could not trust Gen-AI and needed to recheck the feedback it generated: *“I don’t think the Gen-AI is perfect or Gen-AI is always correct, so I need to check again” (I7)* due to potential mistakes or hallucinations. The answers Gen-AI provided were seen as too broad and generic: *“And then he gives like a long paragraph of text and the Gen-AI is completely useless. It has like vague and general language that cannot be used whatsoever” (I9).* The reason for this was that Gen-AI lacks creative thinking: *“it has seen thousands of papers in that language. Just your specific content, it will not have seen, so it will not know” (I9).*

Ethics in the context of group work

Participants highlighted that it was necessary to mention limitations of Gen-AI use while discussing the group strategy because *“basically, the texts are the generative AI use, like the text from like just random documents for learning. So, it might result in like you actually using others work without noticing” (I1).* Another important ethical point was that, in individual pieces of writing created with the help of Gen-AI within the group task, there should not be any false information: *“Everyone can use in their, in their way. So whatever they want, they can, like, do anything. And, like,*

except generating or writing false information” (I8). Taken together, these findings suggest that students viewed Gen-AI as a valuable collaborative writing support, while being aware of its limitations, potential risks for learning, and ethical challenges.

Discussion & conclusion

This study explored the strategies that a sample of graduate students employed when interacting with Gen-AI during the preparation of group writing assignments. The findings reveal that these strategies consisted of three key themes: group strategy, strategy characteristics, and strategy considerations, each of which shape how students engage with their own work, their peers’ contributions, and Gen-AI tools. These insights extend prior research on Gen-AI use, building on studies that examined its role in individual [33,19] and group writing [25,26]. Importantly, the study addresses an underexplored area of Gen-AI use in collaborative writing, which remains less understood than individual student–AI interaction despite the growing adoption of Gen-AI in group learning activities. By framing these results within existing literature, this study contributes to a growing understanding of how Gen-AI transforms collaborative writing practices in specific educational contexts. Methodologically, the study contributes by combining interview data, written reflections, and collaborative writing documents, allowing reported Gen-AI practices to be compared with documented evidence of how these practices were enacted during collaborative writing.

It is important to note that our findings should be interpreted in light of the specific context of the study, which involved a small group of graduate students majoring in AI research with prior experience using AI tools and, therefore, may not directly generalize to broader student populations. At the same time, this context provides insight into more advanced patterns of Gen-AI use that may not yet be observable among less experienced users. Given the qualitative and exploratory nature of the study, the identified themes should be interpreted as context-specific rather than as a comprehensive taxonomy of collaborative Gen-AI use.

The first theme, group work, pertains to whether students established a collective strategy and how they engaged with each other’s contributions during assignment preparation. In many cases, students collaboratively decided on a strategy and adhered to it; however, there were instances where an individual suggested implementing a strategy, aligning with Yim et al.’s [32] observations of leadership dynamics within group settings. Another critical aspect of group work involved how students interacted with their peers’ contributions. Some monitored each other’s progress online, while others remained largely indifferent to their peers’ work. This mirrors prior findings that described students focusing exclusively on their assigned sections of a task rather than adopting a more integrated approach [38,32]. When students focused solely on their individual contributions, Gen-AI proved to be helpful in merging individual drafts during the final stages of the assignment. This use of Gen-AI to integrate separate sections into a cohesive product represents a new function of group use of Gen-AI that has received limited attention in existing research. From a collaborative perspective, this finding suggests that Gen-AI may support coordination in groups where students work independently by helping to integrate distributed contributions into a shared product. At the same time, such use may reduce the need for students to engage in negotiating and revising each other’s ideas.

The second theme of Gen-AI use strategy characteristics pertains to the specific mechanics of interacting with this technology. The findings revealed that Gen-AI in the group tasks was used for technical tasks, for example, to enhance the language of writing and facilitate translation, aligning with prior research on individual [17,19] and group use of Gen-AI [25,26]. Non-technical functions of Gen-AI, such as generating ideas and creating drafts or outlines, are also consistent with previous studies on both individual [15,33] and group [11,26] uses of this tool. The task of merging individual drafts into one product extends the individual-level use of Gen-AI for draft refinement described by [15] to a

group context and aligns with Lowry et al.'s [23] taxonomy of parallel writing followed by integration. This group-level use of Gen-AI to work with final drafts addresses the challenge of synthesizing individual contributions into a cohesive group product, as noted by Buyserie [34]. Another notable finding was the distinction between AI-first and human-first writing approaches. This distinction is analytically useful because it highlights differences in how responsibility for idea generation and text creation is distributed between students and the technology during collaborative writing. These findings indicate that Gen-AI may affect not only the quality of individual contributions but also the way collaborative writing is organized, potentially reshaping how responsibility for drafting, revision, and integration is distributed among group members.

The third theme of Gen-AI use strategies considerations reveals how different perceptions and concerns affect group strategies and mechanics of Gen-AI application. It was found that these considerations include benefits of using Gen-AI. Participants reported that Gen-AI was helpful in making the writing process faster and easier and assisted in enhancing the quality of the final product. Additionally, it was reported that Gen-AI contributed to students' long-term development by supporting their skill development and meaningful learning over time. Students highlighted such potential drawbacks of Gen-AI as the risk of generating false information and skill deterioration. When talking about ethical aspects of using Gen-AI, students highlighted their concerns about the accuracy of their peers' AI-generated contributions and the risk of plagiarism if the peers used AI outputs without verification. While prior work has described students' concerns about negative effects of Gen-AI usage on their own learning and writing quality, the present findings illustrate that the effects (both positive and negative) of Gen-AI usage in group work extend beyond the individual to the whole group as well. This suggests that decisions about Gen-AI use in collaborative writing cannot be viewed solely as individual choices, as the benefits, limitations, and risks associated with Gen-AI may affect other group members and the collaborative process as a whole.

Taken together, these findings have broader implications for collaborative learning and the collaborative construction of knowledge. Previous research has conceptualized collaborative writing as a process of shared understanding development through discussion, negotiation, and the integration of individual contributions [24,23]. In the present study, these processes remain collaborative but are increasingly mediated by Gen-AI. Students often used Gen-AI to produce initial drafts, refine their individual contributions, and integrate them into one coherent document. Consequently, some interactions that traditionally occurred between group members shifted toward interactions between individual students and Gen-AI. This finding extends existing descriptions of collaborative writing by showing that certain writing functions may no longer be carried out exclusively among human collaborators but can also be distributed across individual students and Gen-AI.

Collaborative learning suggests that explaining ideas, negotiating meaning, and responding to alternative perspectives are mechanisms through which learners co-construct knowledge (Dillenbourg, 1999; [21]). Our findings suggest that when Gen-AI performs part of the drafting or integration process, groups may complete writing tasks more efficiently, but opportunities for these collaborative processes may be reduced. In particular, using Gen-AI to integrate independently written sections may reduce the need for students to discuss inconsistencies and reconcile alternative viewpoints when working on the group product. Because these interactions encourage learners to explain, justify, and refine their thinking together, reducing them may shift some opportunities for collaborative knowledge construction from peer-to-peer interaction toward student-AI interaction. At the same time, our findings indicate that in some cases students evaluated, revised, and discussed AI-generated outputs before incorporating them into the group document. This suggests that Gen-AI can support collaborative knowledge construction when its outputs become the subject of discussion

instead of being accepted without further interaction.

Limitations

The study has several limitations. The small sample size of 10 participants limits the generalizability of the findings. Additionally, the sample was homogeneous, as all the participants were students of the Graduate School of AI. As a result, the participants were likely to possess above-average experience with Gen-AI compared to the broader student population, which may limit the transferability of the findings to other educational contexts. While this purposive sampling was employed to ensure that all participants had a high degree of familiarity with the use of Gen-AI tools in writing, this method potentially restricted the diversity of perspectives and experiences. The study was conducted at a single university, which further narrows the contextual scope and raises questions about the broader applicability of identified strategies. Third, the findings were derived from self-reported data and participant interviews. The lack of direct observations of group dynamics or real-time strategy development limits the ability to objectively assess how these strategies were implemented. Consequently, the study was unable to capture the real-time development of group strategies and interactions as they occurred during collaborative writing tasks. Furthermore, reliance on self-reported data introduces potential biases, such as social desirability or selective memory, which may affect the accuracy and completeness of the findings.

Future research directions

Future research could involve observing real-time group processes and comparing them with students' self-reported data on Gen-AI use. Experimental studies aimed to assess the outcomes of specific Gen-AI use strategies can also be conducted. Larger-scale studies with cross-disciplinary samples and longitudinal designs could enhance generalizability and track the evolution of Gen-AI use strategies over time. Finally, research on ethical considerations would provide insights into responsible Gen-AI use and its integration into academic practices.

Practical recommendations

To enhance the integration of Gen-AI in collaborative learning, educators should guide students in developing explicit group strategies for Gen-AI use, fostering cohesive teamwork and improving project outcomes, as student groups may not always establish these strategies on their own. Such discussions may include decisions about who will use Gen-AI, for what purposes, and at what stages of the writing process. Tailored guidance on task-specific AI applications should be provided, as Gen-AI usage varies with assignment types. Students may use Gen-AI differently for language-related and content-related aspects of writing, and these uses may require different forms of instructional support. Relatedly, transparency should be encouraged, with students openly discussing their individual and group strategies to promote collaboration and mutual understanding. Educators may further encourage students to reflect on whether they adopt AI-first or human-first writing approaches, as these strategies distribute responsibility for idea generation and text development differently. Finally, particular attention should be paid to the integration stage of collaborative writing, where Gen-AI may help merge individual contributions into a shared text. Reflective practices, such as self-assessment exercises, should be implemented to help students evaluate and refine their strategies over time.

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Ethics approval and consent to participate and publication

This research was approved by an IRB at KAIST in April 2024. Approval number: KH2024–091 Participants in this research gave their consent to be a part of the project, as well as to have their data collected and analyzed.

Participants consented to having their data used after anonymisation.

Data statement

The data supporting this research is available from corresponding author

AI usage declaration

We used generative AI tools (ChatGPT-5.5, Perplexity) to check

Appendix A

Fig. A1, Fig. A2, Fig. A3

abilities, such as English lectures. In addition, there exists considerable room for improvement for robot surrogates, as studies to enhance local agents' language understanding abilities are actively being conducted and yielding promising outcomes. Nevertheless, the observation that even the best-performing robot surrogate could not replace direct student participation indicates that actual attendance remains the core part of education. Further studies incorporating robots not as surrogates but as agents to provide supplementary support may offer more effective solutions for real-world classrooms.

Fig. A1. Example of Gen-AI-assisted stylistic enhancement in group writing.

Nevertheless, the main reason KAIST students buy ramen is for time efficiency. Students already know that healthy food is better, so if healthy food is still slow or not easily cooked, it cannot serve as an alternative to instant food. Furthermore, Healthy foods are difficult to sell in stores and require cooking, making them difficult to be consumed late at night. Considering that instant food is often consumed in places other than the cafeteria, such as dormitories, it is hard to effectively reduce instant food consumption.

Fig. A2. Illustration of Gen-AI Use for translation in a group writing document.

Comparison to previous works

Our results are consistent with existing literature on the effectiveness of robotic surrogates. Previous studies have highlighted the potential of such devices to relieve students of attendance burdens without significantly affecting their grades (Johnson & Lee, 2021). Moreover, iStooge's superior performance, especially in comparison to HUBO and T9000, underscores the importance of advanced communication skills to handle common situations and technical capabilities (Chen et al., 2019). This supports the broader literature suggesting that the balance between technical proficiency and communication skills is crucial for surrogate robots in educational settings.

Fig. A3. Use of Gen-AI for revising the draft in a group writing task.

grammar and improve clarity and flow of some parts of the text

CRediT authorship contribution statement

Anna Korchak: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Jamie Costley:** Writing – review & editing, Validation, Methodology, Funding acquisition, Conceptualization. **Mik Fanguy:** Writing – review & editing, Validation, Methodology, Investigation, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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